Matrix and Tensor Convolutions

Erisa Terolli

Slide Credits to Shusen Wang

Inner Products

Vector Inner Product

- $\mathbf{a} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 5 \\ 6 \\ 7 \\ 8 \end{bmatrix}$.

- Inner product:

$$\langle \mathbf{a}, \mathbf{b} \rangle = \mathbf{a}^T \mathbf{b} = \sum_i a_i b_i = 70.$$

Matrix Inner Product

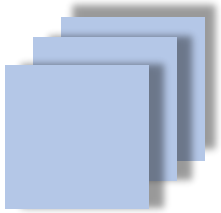
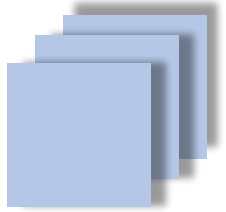
- $\mathbf{A} = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 5 & 7 \\ 6 & 8 \end{bmatrix}$.

- Inner product:

$$\langle \mathbf{A}, \mathbf{B} \rangle = \sum_i \sum_j a_{ij} b_{ij} = 70.$$

- Property: $\langle \mathbf{A}, \mathbf{B} \rangle = \langle \mathrm{vec}(\mathbf{A}), \mathrm{vec}(\mathbf{B}) \rangle$.

Tensor Inner Product

- $\mathbf{A} =$  and $\mathbf{B} =$  .

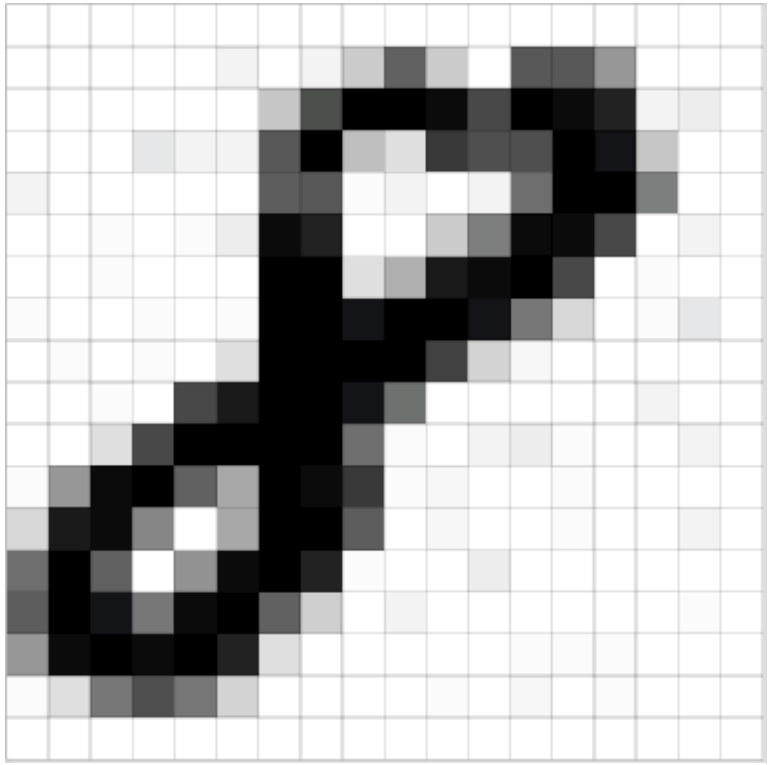
- Inner product:

$$\langle \mathbf{A}, \mathbf{B} \rangle = \sum_i \sum_j \sum_k a_{ijk} b_{ijk} .$$

- Property: $\langle \mathbf{A}, \mathbf{B} \rangle = \langle \mathrm{vec}(\mathbf{A}), \mathrm{vec}(\mathbf{B}) \rangle$.

Convolution for Images

An Image is A Matrix/Tensor of Pixel Values



The MNIST Dataset

- Each grayscale image, $\mathbf{x}_j$, is a 28×28 matrix.

- (Previously, we reshape it to a 784-dim vector.)

Convolution

Input Image

5×5

1	1	1	0	0
0	1	1	1	0
0	0	1	1	1
0	0	1	1	0
0	1	1	0	0

Filter (Kernel)

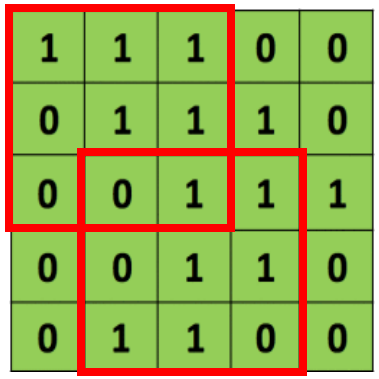
3×3

1	0	1
0	1	0
1	0	1

Convolution

Input Image

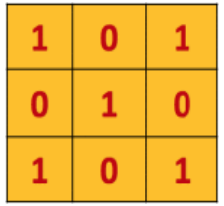
5×5



3×3 patch

Filter (Kernel)

3×3



Question: how many 3×3 patches does a 5×5 image have?

Answer: $(5 - 3 + 1) \times (5 - 3 + 1) = 9.$

Convolution

Input Image

5×5

1	1	1	0	0
0	1	1	1	0
0	0	1	1	1
0	0	1	1	0
0	1	1	0	0

Filter (Kernel)

3×3

1	0	1
0	1	0
1	0	1

Convolution:

4	3	4
2	4	3
2	3	4

Result

3×3

Convolution

Input Image

5×5

1	1	1	0	0
0	1	1	1	0
0	0	1	1	1
0	0	1	1	0
0	1	1	0	0

Filter (Kernel)

3×3

1	0	1
0	1	0
1	0	1

Convolution:

4	3	4
2	4	3
2	3	4

Result

3×3

The value 4 is the inner product of the patch

1	1	1
0	1	1
0	0	1

and the filter

1	0	1
0	1	0
1	0	1

Convolution

Input Image

5×5

1	1	1	0	0
0	1	1	1	0
0	0	1	1	1
0	0	1	1	0
0	1	1	0	0

Filter (Kernel)

3×3

1	0	1
0	1	0
1	0	1

Convolution:

4	3	4
2	4	3
2	3	4

Result

3×3

The value 3 is the inner product of the patch

1	1	0
1	1	1
0	1	1

and the filter

1	0	1
0	1	0
1	0	1

.

Convolution

Input Image

5×5

1	1	1	0	0
0	1	1	1	0
0	0	1	1	1
0	0	1	1	0
0	1	1	0	0

Filter (Kernel)

3×3

1	0	1
0	1	0
1	0	1

Convolution:

4	3	4
2	4	3
2	3	4

Result

3×3

The value 4 is the inner product of the patch

1	1	1
1	1	0
1	0	0

and the filter

1	0	1
0	1	0
1	0	1

.

Convolution

Input Image

5×5

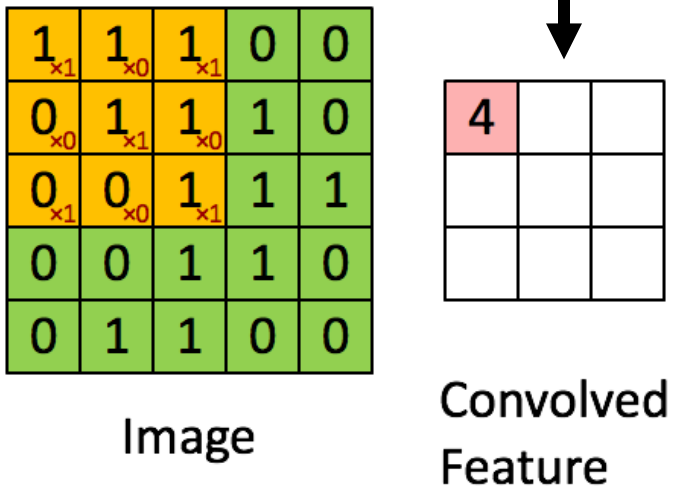
1	1	1	0	0
0	1	1	1	0
0	0	1	1	1
0	0	1	1	0
0	1	1	0	0

Filter (Kernel)

3×3

1	0	1
0	1	0
1	0	1

Convolution:



Image

Convolved Feature

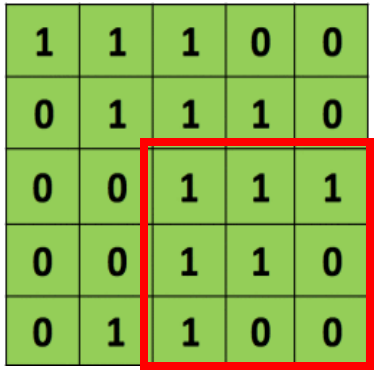
Result

3×3

Convolution

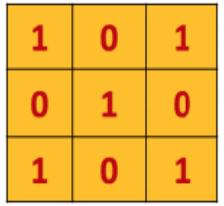
Input Image

5×5

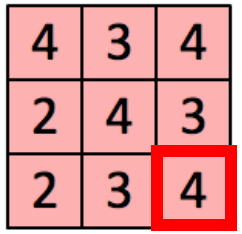


Filter (Kernel)

3×3



Convolution:



Result

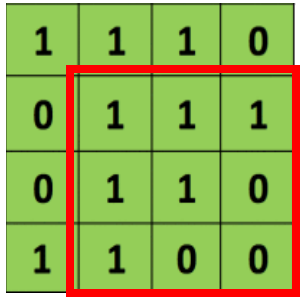
3×3

Question: Why is the result a 3×3 **matrix?**

Convolution

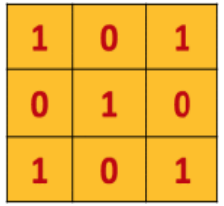
Input Image

4×4

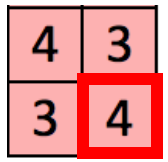


Filter (Kernel)

3×3



Convolution:



Result

2×2

Convolution

Input Image

5×5

1	1	1	0	0
0	1	1	1	0
0	0	1	1	1
0	0	1	1	0
0	1	1	0	0

Filter (Kernel)

3×3

1	0	1
0	1	0
1	0	1

Convolution:

4	3	4
2	4	3
2	3	4

Dimensions:

- Input: $d_1 \times d_2$
- Filter: $k_1 \times k_2$
- Output: $(d_1 - k_1 + 1) \times (d_2 - k_2 + 1)$

Convolution

Input Image



	Filters	**Outputs**
Identity	$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$	

Convolution

Input Image	Filters	Outputs
	Edge detection $\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$	

- The inner product of $\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$ and $\begin{bmatrix} 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 \end{bmatrix}$ is 0.
- The inner product of $\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$ and $\begin{bmatrix} 0.1 & 0.1 & 0.1 \\ 0.1 & 0.2 & 0.1 \\ 0.1 & 0.1 & 0.1 \end{bmatrix}$ is 0.8.

Convolution

Input Image



	Filters	**Outputs**
Sharpen	$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$	

- The inner product of $\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 \end{bmatrix}$ is 0.1.
- The inner product of $\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0.1 & 0.1 & 0.1 \\ 0.1 & 0.2 & 0.1 \\ 0.1 & 0.1 & 0.1 \end{bmatrix}$ is 0.6.

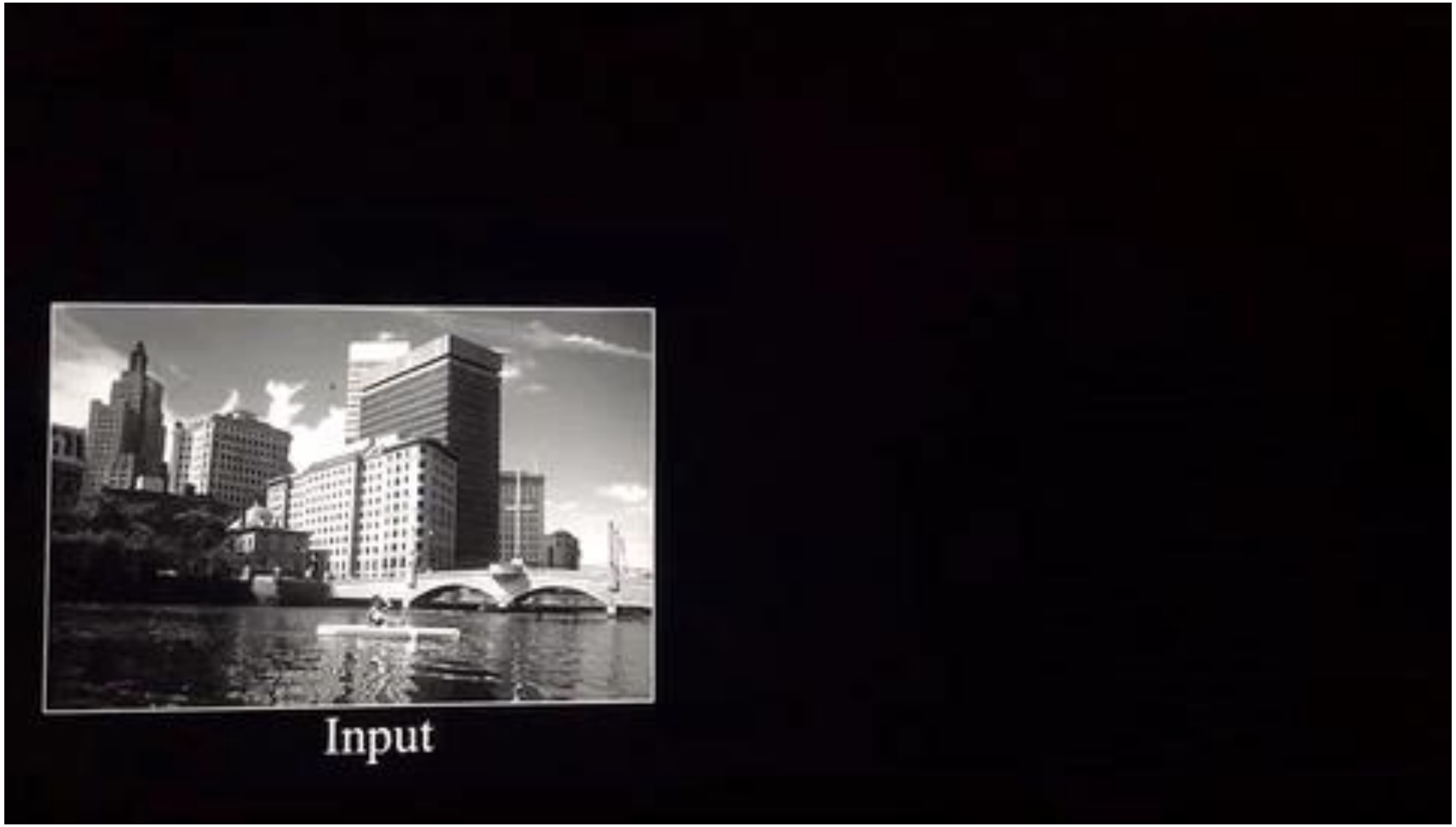
Convolution

Input Image



	Filters	Outputs (or feature map)
Box blur (normalized)	$\frac{1}{9}\begin{bmatrix}1 & 1 & 1\\1 & 1 & 1\\1 & 1 & 1\end{bmatrix}$	
Gaussian blur (approximation)	$\frac{1}{16}\begin{bmatrix}1 & 2 & 1\\2 & 4 & 2\\1 & 2 & 1\end{bmatrix}$	

Convolution



Convolution: Zero Padding

Input Image

5×5

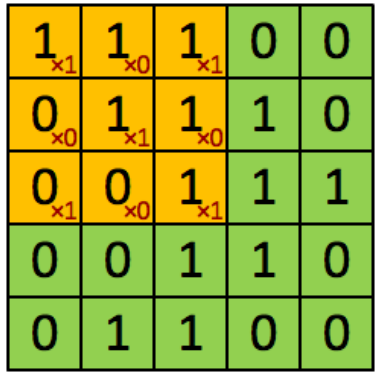
1	1	1	0	0
0	1	1	1	0
0	0	1	1	1
0	0	1	1	0
0	1	1	0	0

Filter (Kernel)

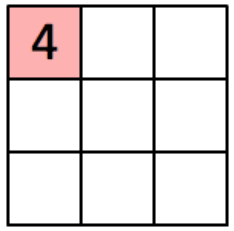
3×3

1	0	1
0	1	0
1	0	1

Convolution:



Image



Convolved Feature

Dimensions:

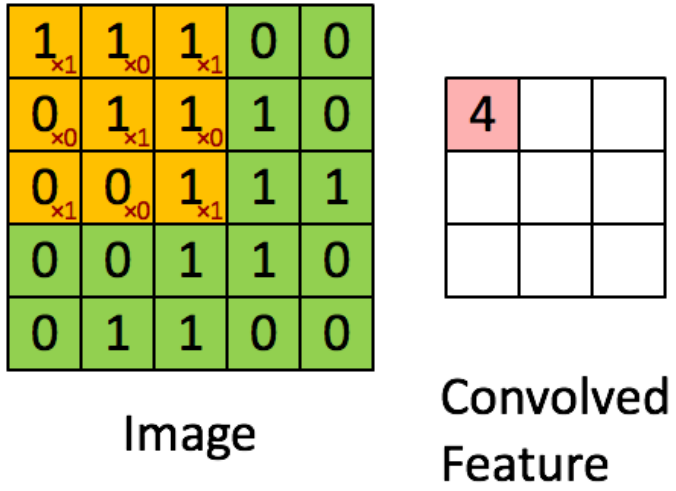
- Input: $d_1 \times d_2$
- Filter: 3×3
- Output: $(d_1 - 2) \times (d_2 - 2)$

Convolution: Zero Padding

Problem: the output is smaller than the input!

- If the input is 28×28 and the filter is 3×3, you can have at most 13 convolutional layers. (The output of the 13th layer is 2×2.)

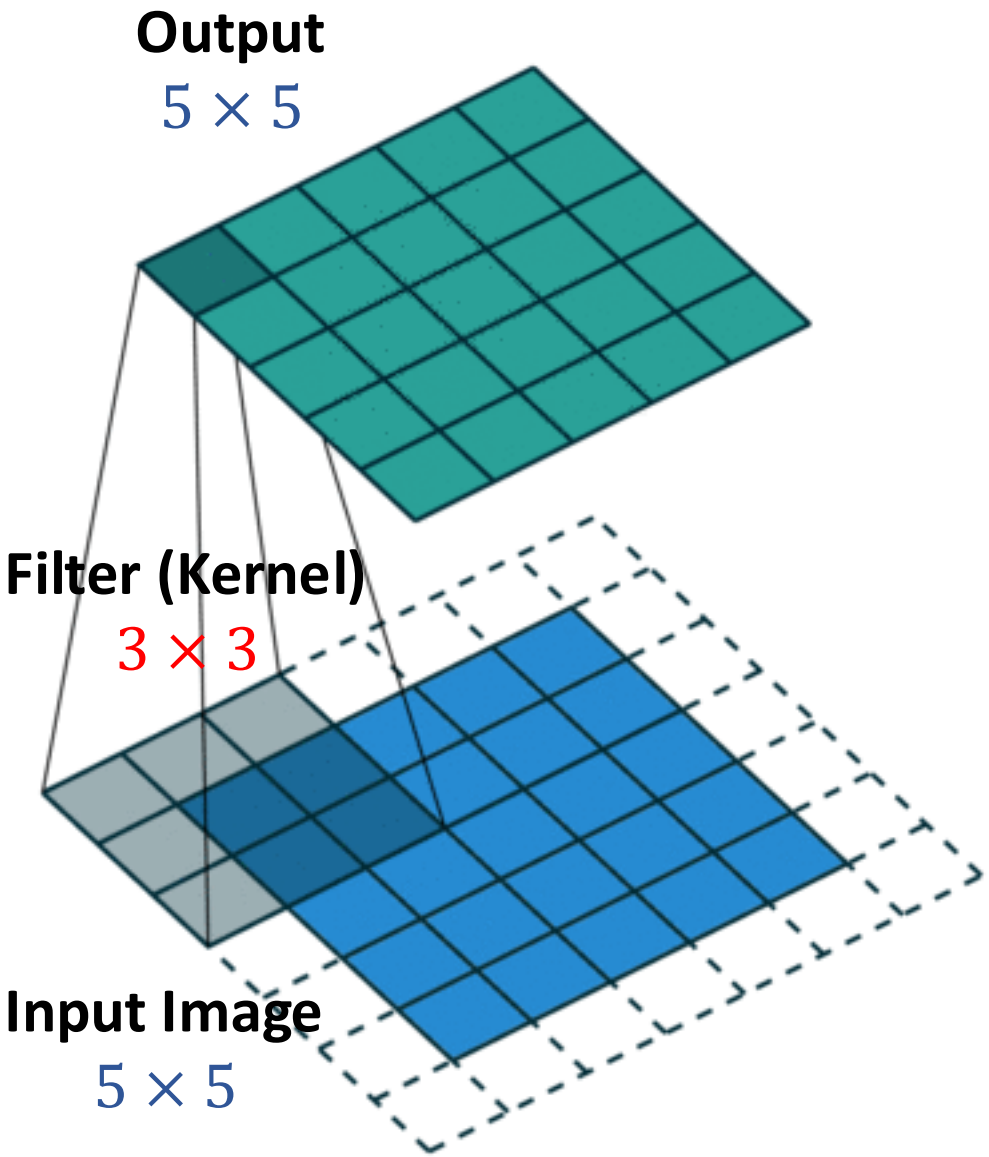
Convolution:



Dimensions:

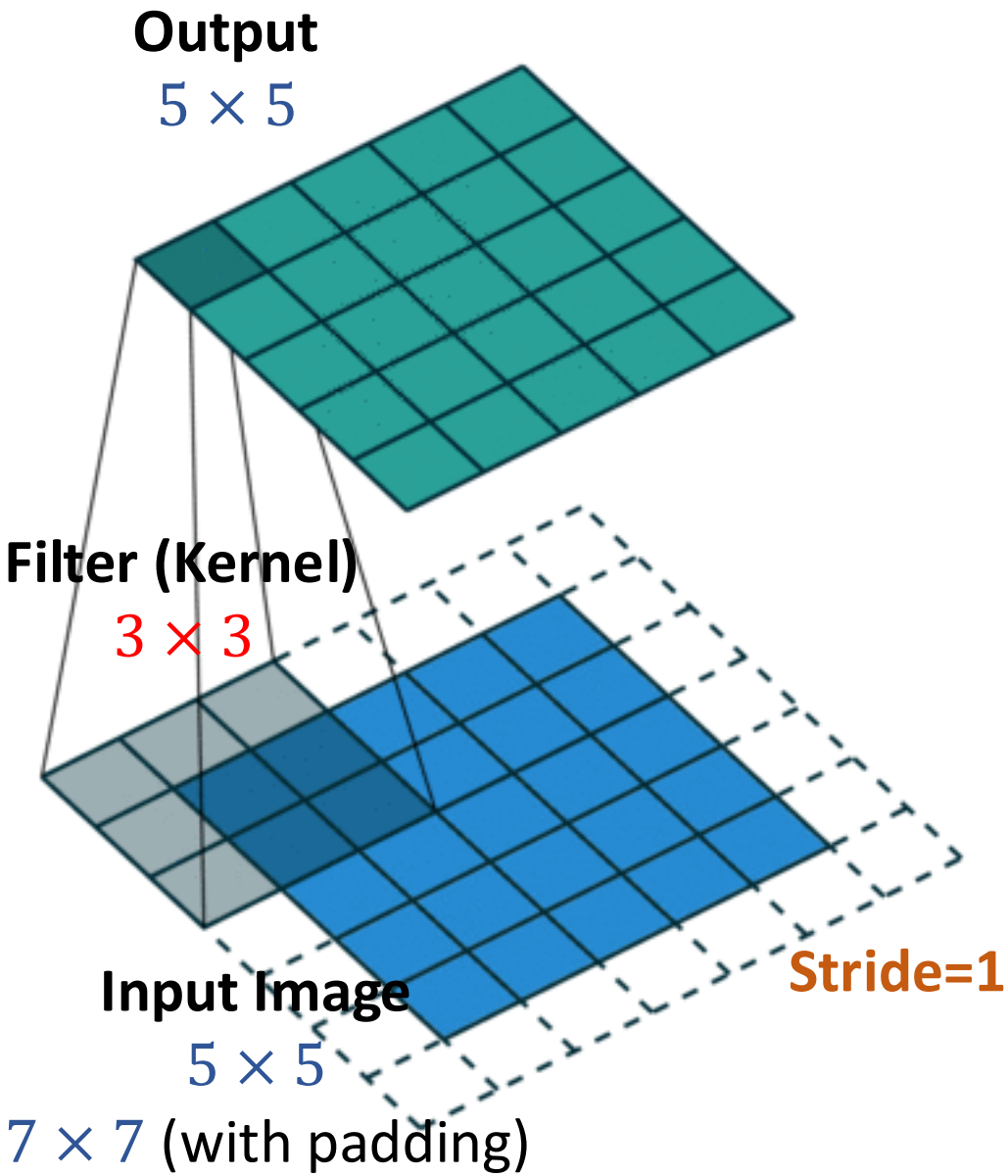
- Input: $d_1 \times d_2$
- Filter: 3×3
- Output: $(d_1 - 2) \times (d_2 - 2)$

Convolution: Zero Padding



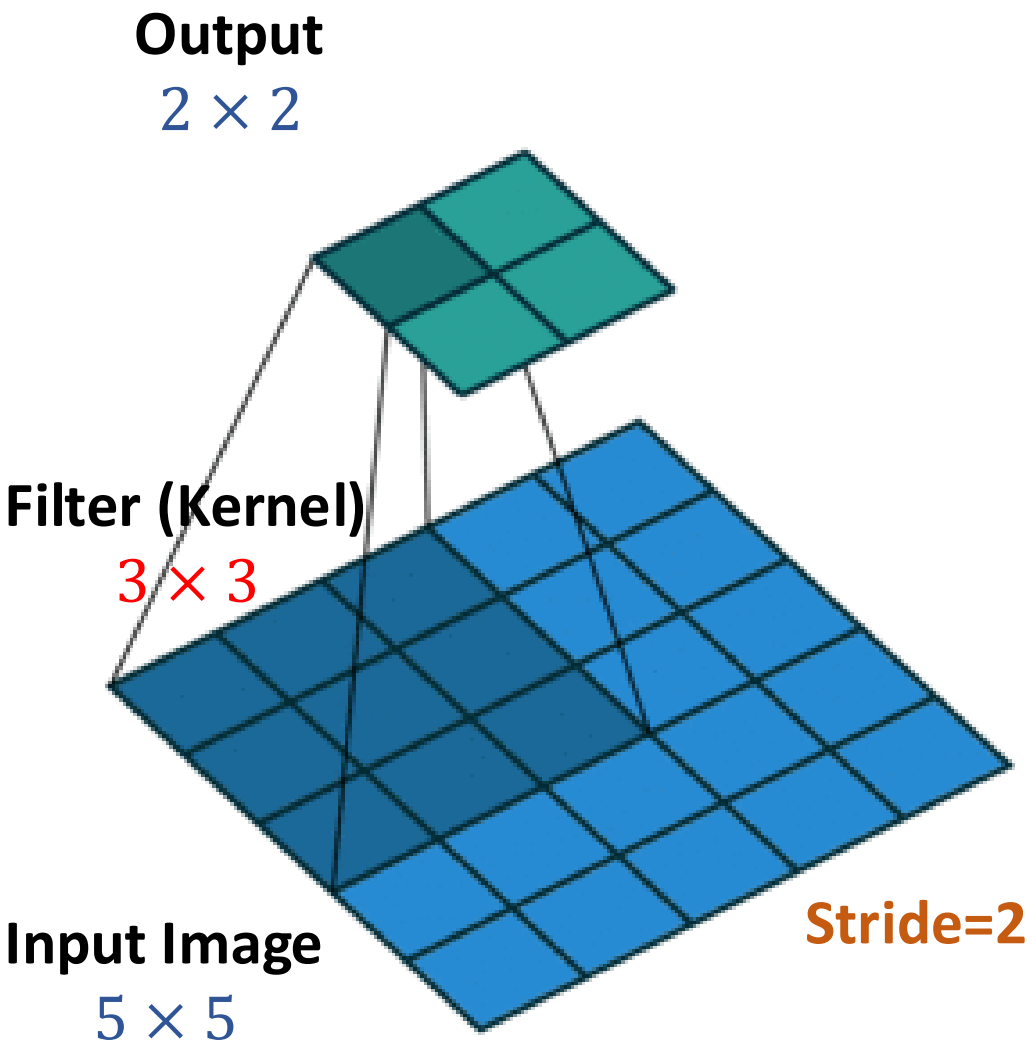
- Add a “boarder” of all-zeros.
- Increase the input shape:
  - From $d_1 \times d_2$ to $(d_1 + 2) \times (d_2 + 2)$.
- If the filter is 3×3, the output is $d_1 \times d_2$.

Convolution: Stride



- In the previous examples, the stride is 1.
  - The filter moves 1 step each time.

Convolution: Stride

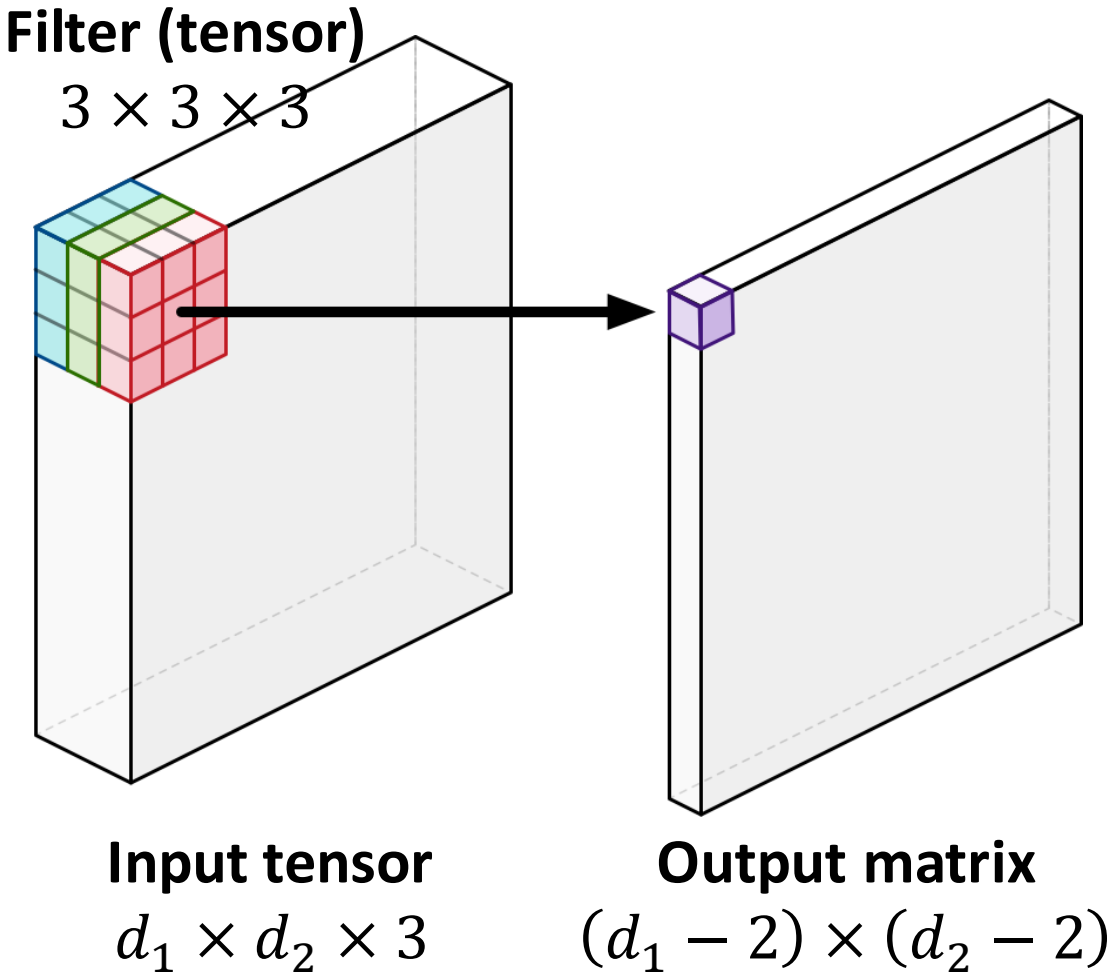


- In the previous examples, the stride is 1.
  - The filter moves 1 step each time.
- The stride can be 2 or even larger.

Dimensions:

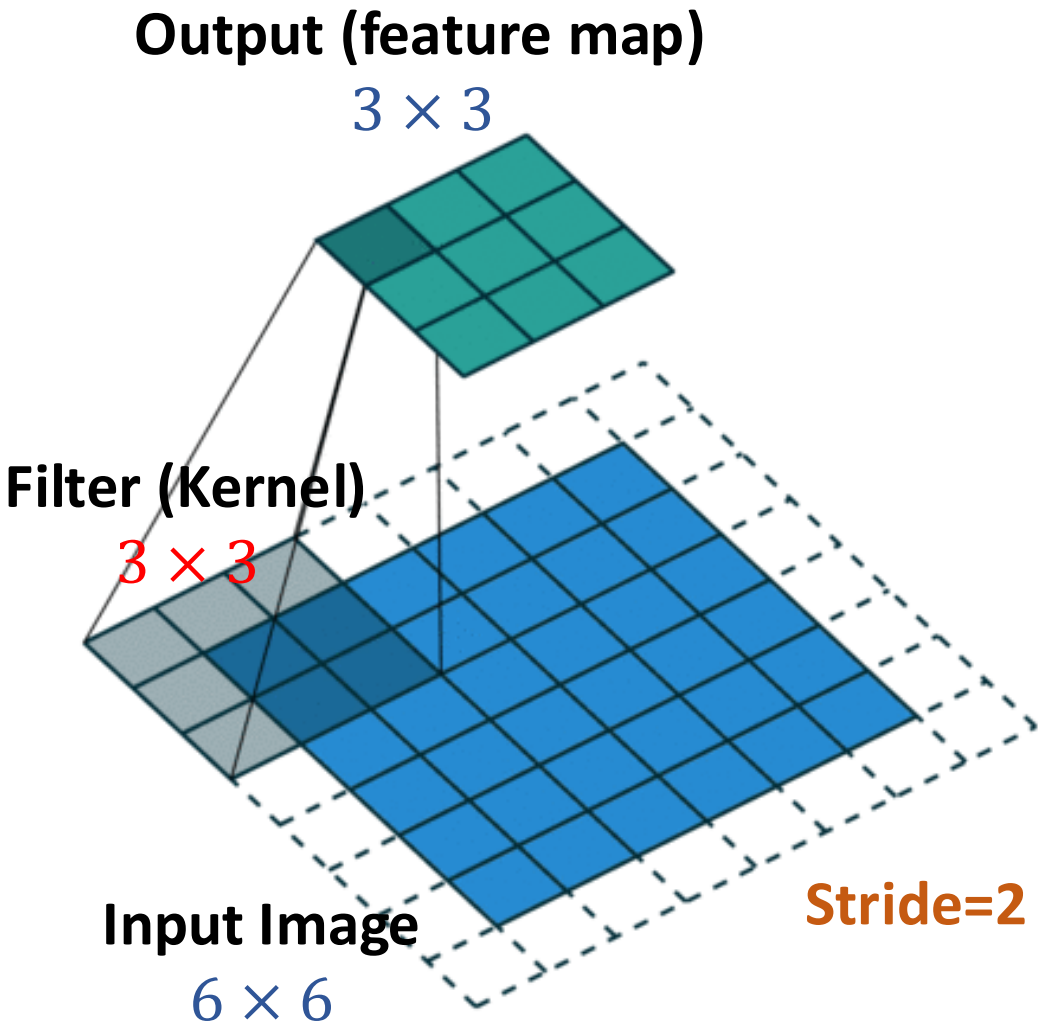
- Input: $d_1 \times d_2$
- Filter: $k_1 \times k_2$
- Stride: s
- Output: $\left(\left\lfloor \frac{d_1 - k_1}{s} \right\rfloor + 1\right) \times \left(\left\lfloor \frac{d_2 - k_2}{s} \right\rfloor + 1\right)$

Convolution for Tensors



- The input is the order-3 tensor of shape $d_1 \times d_2 \times 3$, e.g., a color image.
- The filter is a order-3 tensor of shape $3 \times 3 \times 3$.
- The output is a matrix (order-2 tensor) of shape $(d_1 - 2) \times (d_2 - 2)$.

Convolution: Key Concepts



- Input, filter (kernel), and output (feature map).
- Convolution for matrix (order-2 tensor).
- Convolution for order-3 tensor.
- Zero padding.
- Stride.

Thank you!